

Introduction to Quantum Computing

Part I Linear Algebra for Quantum Computing

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Chapter 1:

What is Quantum Computing?

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What is Quantum Computing?

- In a classical computer, a **bit** is represented by a physical quantity with two states to represent 0 and 1.
 - It can be the voltage in a circuit (states, 5 V for 1 and 0 V for 0).
 - It can be the magnetization direction in a magnetic drive (e.g. up for 1 and down for 0).
 - It can be the charge density in the charge storage layer in the flash memory (e.g. low for 1 and high for 0).
- In a quantum computer, the information is represented by **qubits**.
- Each qubit still has two distinct states, “0” and “1” and they are **orthogonal** to each other because they cannot exist with 100% certainty at the same time.
- We also call them the **basis vectors** or **basis states**.
- A qubit may also have many other states which are the **superposition** of the two basis states.
- The two orthogonal basis states are allowed to exist at the same time with <100% certainty.

What is Quantum Computing?

- Superposition is one of the fundamental operation principles in quantum computing.
- Superposition allows us to perform a calculation on multiple basis states at the same time.
- We learned superposition in wave theory in which two wavelets can superimpose on each other.
- In electromagnetism, light can be polarized in any direction. Any linear polarization is just a linear combination of horizontal and vertical polarization.
- If we represent the horizontal polarization as a horizontal vector and the vertical polarization as a vertical vector, any linear polarization, $\vec{V}$ is just a linear combination of the two vectors.

$$\vec{V} = \alpha \hat{V}_x + \beta \hat{V}_y$$

- where α and β are just some numbers (coefficients) and $\hat{V}_x$ and $\hat{V}_y$ are the horizontal and vertical polarization unit vectors, respectively.
- We can write in the **Dirac bra–ket notation**:

$$|V\rangle = \alpha|V_x\rangle + \beta|V_y\rangle$$

What is Quantum Computing?

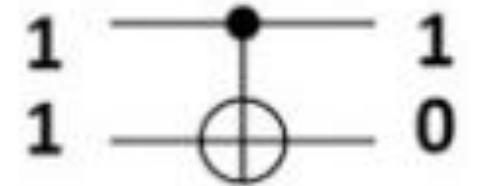
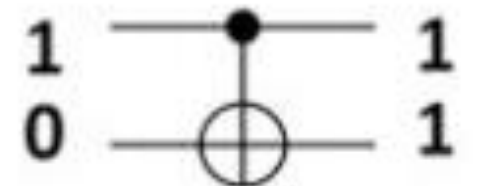
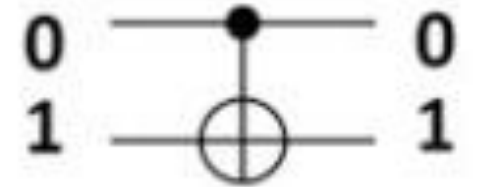
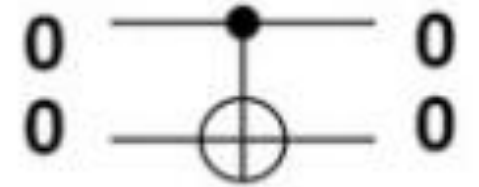
	Classical Computing	Quantum Computing
Representation of Information	0, 1, (<i>bit</i>) (voltage level)	$a 0\rangle + b 1\rangle$ (<i>qubit</i>) (superposition of basis states)
Operation Principles	Classical Boolean Algebra	Superposition, Entanglement, Interference
Logic Gate	Usually, non-reversible	Unitary, reversible
Robustness	Very robust, immune to noise even at 400K	Very sensitive to noise, operating at mK, low fault tolerance.
Measurement	Can copy, deterministic	No-cloning, statistical
Hardware	Transistor	Trapped Ions, Josephson Junction, Topological Insulator, Photonics, Defect Center, Electron spin, Neutral Atoms,
Information size	Process 1 n-bit state at a time	May process 2^n n-bit state at a time

What is Quantum Computing?

- Classical gates (left) are usually non-reversible, but quantum gates (right) have 1-to-1 correspondence between the output and input.



NAND Gate



CNOT Gate

What is Quantum Computing?

- **Reversible Computing:**

- In classical computing, logic gates are non-reversible.
- In quantum computing, every operation has to be reversible (except measurement).
- Quantum mechanics requires nature to evolve in a “unitary” way, and as a result, every quantum gate has to be unitary and thus reversible.
- We can also regard every quantum gate as a rotation operator for a vector.

- **Robustness:**

- A classical computer is very robust against noise. We can have billions of transistors working well even at 400 K.
- However, a quantum computer is very sensitive to noise. Most of them need to work in the mK range. 1 mK is 1/1000 K.
- Liquid helium (L-He) temperature is 4.2 K. This means that the quantum computer chip needs to operate in an environment a hundred times cooler than the L-He.
- This is because the thermal noise is smaller at a lower temperature. Thermal noise is proportional to temperature.
- Qubits are usually implemented with a physical qubit with small state energy separation (e.g. the separation of a spin qubit's up and down is about 0.2 meV).
- Therefore, low temperature is needed so that the thermal energy (noise) is much less than the state separation energy.
- To further mitigate the impact of noise, error correction and signal processing are very important in quantum computing.

What is Quantum Computing?

- **Measurement and Collapse:**

- For a classical measurement, you still copy it to a register or memory.
- In quantum computing, there is a theory called the **No Cloning Theory**. It prevents us from cloning all quantum states.
- Therefore, we cannot copy a quantum state from one place to another place.
- We can also regard every quantum gate as a rotation operator for a vector.
- **Teleportation**, it seems that we can “copy” something from one place to another **in no time**.
- But in **quantum teleportation**, when we “copy” a state, we need to destroy the state at the source. So actually, it is not a copy action and that is why it is called teleportation.
- When measure a superposition state, the state will collapse to one of the basis states (0 or 1) and this is completely random.
- Then how a quantum computer is going to be useful if even the measurement is random?
- In some algorithms, we design it so that eventually it will give us a certain basis state for sure.

- **Implementation of Qubits:**

- In a quantum computer, we need something small so that quantum phenomena can exhibit.
- It also needs to have two distinguishable quantum states as the basis states (0 and 1).
- We can use spin up to represent 0 and spin down to represent 1.
- According to quantum mechanics, the electron can be in a superposition of spin up and spin down.
- Therefore, the spin of an electron under an external magnetic field can be used to implement a qubit.
- There are other possible qubit implementations such as trapped ions, superconducting qubit using Josephson Junction, topological insulator, photons, defect centers in diamond or silicon carbide.
- There are trade-offs between them such as the size of the device, the speed, the coherence time, and scalability.

What is Quantum Computing?

- **Information Size:**

- The power of superposition is the amount of information that can be stored when we have more than 1 qubit.
- How many possible states can we store with n physical bits in a classical computer?
- It is 2^n states. Of course, we can only store one of them at a time. For example, with 2 physical bits, we can store either the state 3 (11 in binary), 2 (10 in binary), 1 (01 in binary), or 0 (00 in binary).
- For quantum computers, each of the classical states becomes the basis state and they are orthogonal to each other. So, you have 2^n basis states if you have n qubits.
- The n -qubit system can also be in a superposition state of the 2^n basis states. For example, if there are two photon polarization qubits:

$$|V\rangle = \alpha|V_{x1}V_{x2}\rangle + \beta|V_{x1}V_{y2}\rangle + \gamma|V_{y1}V_{x2}\rangle + \delta|V_{y1}V_{y2}\rangle$$

- where α , β , γ , and δ are just some numbers and V_{x1} , V_{y1} , V_{x2} , and V_{y2} are the horizontal and vertical polarization vectors of the first and the second photons, respectively.
- We have four basis vectors now, $|V_{x1}V_{x2}\rangle$, $|V_{x1}V_{y2}\rangle$, $|V_{y1}V_{x2}\rangle$, $|V_{y1}V_{y2}\rangle$
- For 100 qubits, we can form a system of 2^{100} basis vectors. If we can process the 100 qubits at the same time, we are processing the 2^{100} coefficients at once. This is the so-called **quantum parallelism**.

Chapter 2:

Basis, Basis Vectors, and Inner Product

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A Romantic Story

- Once upon a time, there was a boy who liked a girl. But he did not know if the girl liked him. The girl treated him patience, but he was not sure if she just treated him as a good friend or she did like him. The boy had heard about quantum mechanics but did not understand too well. For him, the state of the girl was:

$$|State\ of\ the\ girl\rangle = \alpha|Yes\rangle + \beta|No\rangle$$

- Definitely, $|Yes\rangle$ (she likes him) and $|No\rangle$ (she does not like him) are two orthogonal basis vectors that never occur at the same time. For the boy, the girl is in a state which is a superposition of $|Yes\rangle$ and $|No\rangle$.
- The boy dared not ask because asking is a measurement. If it collapses to $|No\rangle$, that would be the end of the world for him.
- After waiting for months, the boy lost his patience, and he decided to ask (which is a measurement by asking “do you like me?”). Unfortunately, the state collapsed to $|No\rangle$! The problem is once the answer is “No,” it is “No” forever.
- Fortunately, the boy did not collapse psychologically. He turned his attention to study and understood quantum mechanics better. He realized that in a quantum system, even if the state has collapsed to a basis state of a certain basis, when you measure in another basis, it will collapse to a basis state of the new basis, which is a superposition of the old basis states.

A Romantic Story

- For him, the old basis was formed by

$$|State\ of\ the\ girl\rangle = \alpha|Yes_L\rangle + \beta|No_L\rangle$$

- where $|Yes_L\rangle$ and $|No_L\rangle$ are the answers to the question “Do you like me?”
- The girl’s state can also be expressed as a superposition of the basis states on another basis.
- For example, the new basis can have the basis states $|Yes\rangle$ and $|No\rangle$ to the question “Do you want to work on a project with me?”. The new basis states can be labeled as $|Yes_P\rangle$ and $|No_P\rangle$
- This is because a girl who does not like him may or may not want to work on a project with him. So $|No_L\rangle$ can coexist with $|Yes_P\rangle$ or $|No_P\rangle$. There are not orthogonal to each other.
- Therefore, after a measurement in the new basis, the state may collapse to $|Yes_P\rangle$ is not orthogonal to $|Yes_L\rangle$ or $|No_L\rangle$, we may have:

$$|Yes_P\rangle = \gamma|Yes_L\rangle + \delta|No_L\rangle$$

- What if he tried again to measure in the old basis, i.e. ask her if she likes him, there is a chance he will get $|Yes_L\rangle$!

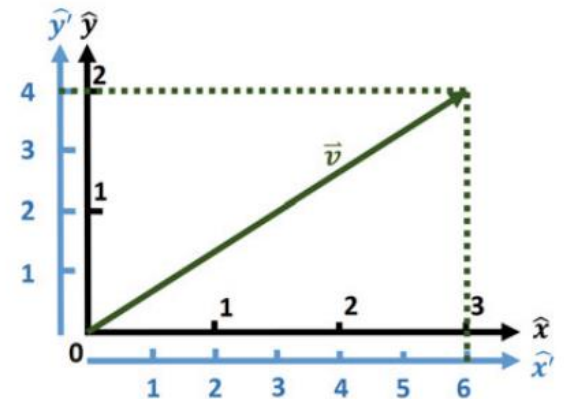
Vectors in Quantum Computing

- Quantum computing is just a manipulation of vectors through operations in the vector space.
- We prepare many vectors in a special way and manipulate them through special quantum gates (operations) and we get the answer we want.
- These operations are also just rotations of vectors in hyperspaces.
- On a 2D plane, we know a vector can be described by its x and y.
- For example, in Pic. 1, we have a vector $\vec{v} = 3\hat{x} + 2\hat{y} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ where $\hat{x}$ and $\hat{y}$ are the unit vectors in the x and y directions. $\hat{x}$ and $\hat{y}$ are just the **basis vectors**.
- In quantum computing, vector and state are the same. We can also write $\vec{v}$ in Dirac's bra-ket notation, as well:

$$|\vec{v}\rangle = 3|\hat{x}\rangle + 2|\hat{y}\rangle$$

- We can also choose another set of basis vectors, $\hat{x}'$ and $\hat{y}'$. $\hat{x}$ and $\hat{y}$ are only half of the lengths of $\hat{x}'$ and $\hat{y}'$, respectively. So that we can represent $\vec{v}$ as follow:

$$|\vec{v}\rangle = 3|\hat{x}\rangle + 2|\hat{y}\rangle = 3(2|\hat{x}'\rangle) + 2(2|\hat{y}'\rangle) = 6|\hat{x}'\rangle + 4|\hat{y}'\rangle = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$



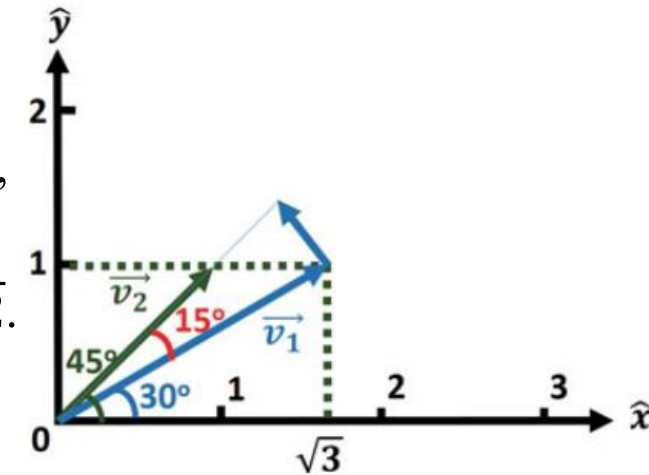
Pic 1: Representations of $|\vec{v}\rangle$ in two different bases.

Vectors Space

- All vectors on a plane can be expressed as a linear combination of the basis vectors $\hat{x}$ and $\hat{y}$ with the form $\vec{v} = 3\hat{x} + 2\hat{y} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$. We say that the basis vectors form or span that vector space.
- For a vector that is out of the plane, it **can't be** expressed as a linear combination of $\hat{x}$ and $\hat{y}$. We say that this out-of-plane vector is in another vector space.
- For example, you bought a dozen eggs. You say you have 12 1-eggs. The basis vector is 1-egg and it spans the whole vector space of eggs.
- Any number of eggs can be represented as a linear combination of 1-egg (here we only have 1 basis vector).
- If you have a dozen apples, you cannot say you have 12 1-eggs because 1-egg does not span the vector space of apples.
- Basically, we are saying we do not have the same vector space and we are not in the same universe.

Inner Products of Vectors

- Vector and state are equivalent. So, we can represent every physical system state as a vector.
- The **inner product** is an important operation of vectors, which is also called the “dot product” or “scalar product”, the physical meaning of the inner product is **projection**.
- Projection is an important concept in quantum mechanics.
- In every measurement, we can say we try to do an operation to project a superposition state to one of the basis states.
- For example, when the boys asked (a measurement) the girl if she liked him and she answered no, the wavefunction is collapsed (projected) into the $|No_L\rangle$ basis state.
- The angle between $\vec{v}_1$ and the x -axis (which has the same direction as $\hat{x}$) is 30° .
- The angle between $\vec{v}_2$ and the x -axis is 45° . So, the angle between $\vec{v}_1$ and $\vec{v}_2$ is 15° .
- From trigonometry, the inner product of the two vectors is $\vec{v}_1 \cdot \vec{v}_2 = |\vec{v}_1| |\vec{v}_2| \cos 15^\circ$, where $|\vec{v}_1|$ and $|\vec{v}_2|$ are the magnitudes (lengths) of $\vec{v}_1$ and $\vec{v}_2$, respectively.
- Using the Pythagorean theorem, $|\vec{v}_1| = \sqrt{\sqrt{3}^2 + 1^2} = 2$ and $|\vec{v}_2| = \sqrt{1^2 + 1^2} = \sqrt{2}$.
- Therefore, $\vec{v}_1 \cdot \vec{v}_2 = |\vec{v}_1| |\vec{v}_2| \cos 15^\circ = 2\sqrt{2} \cos 15^\circ \approx 2.732$. This is the length if we project $\vec{v}_1$ onto $\vec{v}_2$ and scaled by $|\vec{v}_2|$.
- The inner product is telling us how much the two vectors have in common after being scaled by their magnitudes (lengths). We can only use it if the basis vectors are **orthonormal**.



Pic 2: Graphical representations of the inner product of $\vec{v}_1$ and $\vec{v}_2$. The final result needs to be scaled by $|\vec{v}_2|$.

Inner Products of Vectors

- Since we can write all vectors in the column form if the basis vectors are known.
- For example:

$$|\vec{v}_1\rangle = \sqrt{3}|\hat{x}\rangle + 1|\hat{y}\rangle = \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix}$$

$$|\vec{v}_2\rangle = 1|\hat{x}\rangle + 1|\hat{y}\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- To perform the dot product of $\vec{v}_1 \cdot \vec{v}_2$, firstly, we have to rotate the column form of the first vector anti-clockwise by 90° , so that it becomes a **row vector**. And then we can perform matrix multiplication.

$$\vec{v}_1 \cdot \vec{v}_2 = (\sqrt{3} \ 1) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \sqrt{3} \times 1 + 1 \times 1 = \sqrt{3} + 1 \approx 2.732$$

- The answer is exactly the same as $\vec{v}_1 \cdot \vec{v}_2 = |\vec{v}_1| |\vec{v}_2| \cos 15^\circ$.
- In general, if $\vec{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\vec{v}_2 = \begin{pmatrix} c \\ d \end{pmatrix}$, we have $\vec{v}_1 \cdot \vec{v}_2 = (a^* \ b^*) \begin{pmatrix} c \\ d \end{pmatrix} = a^* \times c + b^* \times d$ where a^* and b^* are the **complex conjugate** of a and b , respectively.
- The coefficients in the linear combination of basis vectors can be **complex numbers**.

Higher Dimensional Vector Space

- When there are two basis vectors, it spans a two-dimensional (2D) vector space, just like $\hat{x}$ and $\hat{y}$ span a plane.
- If we add $\hat{z}$, the third basis vector in the height direction, they then span a 3D space.
- If we have n basis vectors, we expect they will span an n -**D** space.
- Any vector $\overrightarrow{v_a}$ must be a linear combination of the n basis vectors.

$$\overrightarrow{v_a} = a_0 \widehat{v_0} + a_1 \widehat{v_1} + a_2 \widehat{v_2} + \cdots + a_{n-1} \widehat{v_{n-1}}$$

where a_0 to a_{n-1} are the n coefficients and $\widehat{v_0}$ to $\widehat{v_{n-1}}$ are the n basis vectors. The coefficients may be complex numbers.

- And for another vector,

$$\overrightarrow{v_b} = b_0 \widehat{v_0} + b_1 \widehat{v_1} + b_2 \widehat{v_2} + \cdots + b_{n-1} \widehat{v_{n-1}}$$

where b_0 to b_{n-1} are the n coefficients which can be real or complex.

- The inner product of the two vectors is

$$\overrightarrow{v_a} \cdot \overrightarrow{v_b} = (a_0^* \ a_1^* \ \dots \ a_{n-1}^*) \begin{pmatrix} b_0 \\ 1 \\ \vdots \\ b_{n-1} \end{pmatrix} = a_0^* \times b_0 + a_1^* \times b_1 + \cdots + a_{n-1}^* \times b_{n-1}$$

Summary

- Vectors are states and vectors are linear combinations of basis vectors.
- The basis vectors can be something very abstract (e.g. spin up) or absurd (1-apple).
- The coefficients in the linear combination can be complex numbers.
- We also reviewed the meaning and methodology of calculating the inner product of two vectors.
- All these concepts apply to higher dimensions.